

Assignment - 1

SE4020 – Mobile Application Design and Development

Semester 1, 2026

Marks – 10%

Deadline – 19th April 2026

You are required to build a Minimum Viable Product (MVP) iOS application based on the **NurseryConnect** case study provided. You will select **exactly two (2) key features** from the case study to implement, and you will adopt **one (1) user role** defined in the case study as the perspective from which your app is built. Your deliverable is an iOS app only — you are **not** required to implement the web application, cloud infrastructure, or any backend services described in the case study. The app should have at least two screens navigated using a navigation controller, incorporate appropriate SwiftUI components, and include a data persistence method of your choice. The scope of the two features you select should be sufficient to represent a meaningful 4-week development effort.

Here are some additional requirements to consider for the assignment:

1. Login, authentication, and access control features must not be included as either of your two selected key features. Your app should launch directly into its main functionality.
2. You must select one (1) user role from the NurseryConnect case study (e.g. Keyworker, Parent/Guardian, Setting Manager, Driver, Catering Staff, or Marketing Coordinator) and build your app entirely from that role's perspective. Clearly state your chosen role at the beginning of your report.
3. You must select exactly two (2) key features from the case study to implement. Features should be chosen to complement each other and create a coherent experience for your selected user role. Justify your feature selection in your report, explaining why these two features are most appropriate for a 4-week MVP.
4. You must submit a written report demonstrating your understanding of the regulatory and compliance requirements outlined in the case study (UK GDPR, EYFS 2024, Ofsted, Children Act 1989, and FSA guidelines as applicable). The report must explain how your iOS app architecture and design decisions comply with, or acknowledge the constraints imposed by, these regulations. You do not need to implement backend compliance mechanisms, but you must show you understand them and reflect them in your design choices.
5. The user interface should be visually appealing, easy to use, and appropriate for the professional childcare context described in the case study. Aim for a professional look and feel that reflects the sensitive nature of the data being handled.
6. The app should be thoroughly tested to ensure it works as expected and is free of bugs.
7. You should provide a detailed write-up explaining your design choices, implementation decisions, and any challenges you faced during the development process.
8. You should also present a demonstration of your app to the class, highlighting its key features and functionality.

Assignment - 1**SE4020 – Mobile Application Design and Development****Semester 1, 2026****Submission Guidelines**

1. This is an individual Assignment.
2. You are expected to design and write code on your own for this assignment. You should not use any code written by anyone else.
3. You are allowed to use any AI generated code for your submission e.g. using GitHub CoPilot, ChatGPT, Gemini, DeepSeek, Claude, Grok, GhostWriter or any similar tools. You are however required to attach detail prompts and responses you have got for the AI Agent.
4. During the viva which we will have, you should be able describe in sufficient details all aspects of your submission including all code and design choices you have made.

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Marking Rubric

Feature Selection & Role Justification | 10 |

- Has the student clearly identified their chosen user role and justified their two selected features? Is the feature selection appropriate for the chosen role and realistic for a 4-week MVP scope?

App Functionality | 20 |

- Functionality (15): Does the app work as intended? Does it incorporate at least two screens with navigation between them? Does it persist data using some method?
- Error Handling (5): Are errors handled gracefully? Are any errors identified and handled appropriately?

User Interface Design | 20 |

- Visual Design (10): Is the visual design of the user interface appealing and consistent? Are appropriate colors, fonts, and layouts used?
- Usability (10): Is the user interface easy to use and navigate? Does it provide clear feedback to the user?

Swift and SwiftUI Knowledge | 20 |

- Code Quality (10): Does the code demonstrate a good understanding of Swift? Is the code well-organized and readable? Are best practices followed, such as naming conventions and separation of concerns?
- UI Components (5): Are SwiftUI components used effectively to build the user interface? Are they customized appropriately?
- Advanced Concepts (5): Does the code demonstrate an understanding of advanced iOS development concepts, such as concurrency, animations, or networking?

Testing and Debugging | 10 |

- Testing (8): Has the app been thoroughly tested, including unit tests and UI tests? Are tests designed to cover all functionality and edge cases?
- Debugging (2): Are any bugs identified and fixed? Are appropriate debugging techniques used?

Documentation | 10 |

- Design (4): Is there a detailed write-up explaining design choices, including user interface design, user experience considerations, and how the design reflects the professional childcare context?
- Implementation (4): Is there a clear explanation of implementation decisions, including the use of third-party libraries or APIs, data persistence strategy, and any constraints or simplifications made for the MVP scope?
- Challenges (2): Are any challenges faced during the development process identified and explained?

Regulatory Compliance Report | 10 |

- Understanding of Regulations (4): Does the report demonstrate a clear and accurate understanding of the relevant regulatory requirements applicable to the NurseryConnect system (UK GDPR, EYFS 2024, Ofsted, Children Act 1989, and FSA guidelines)? Are the key obligations for the student's chosen role and features correctly identified?
- Compliance by Design (4): Does the report explain how the iOS app's architecture, data handling, and UI design decisions reflect and comply with the identified regulatory requirements? Does the student articulate what a full production system would need to implement to achieve complete compliance?
- Depth and Critical Analysis (2): Does the report go beyond surface-level descriptions? Does the student critically analyse any tensions or trade-offs between regulatory compliance and usability, and propose thoughtful mitigations?